CPU-GPU with Unified Memory

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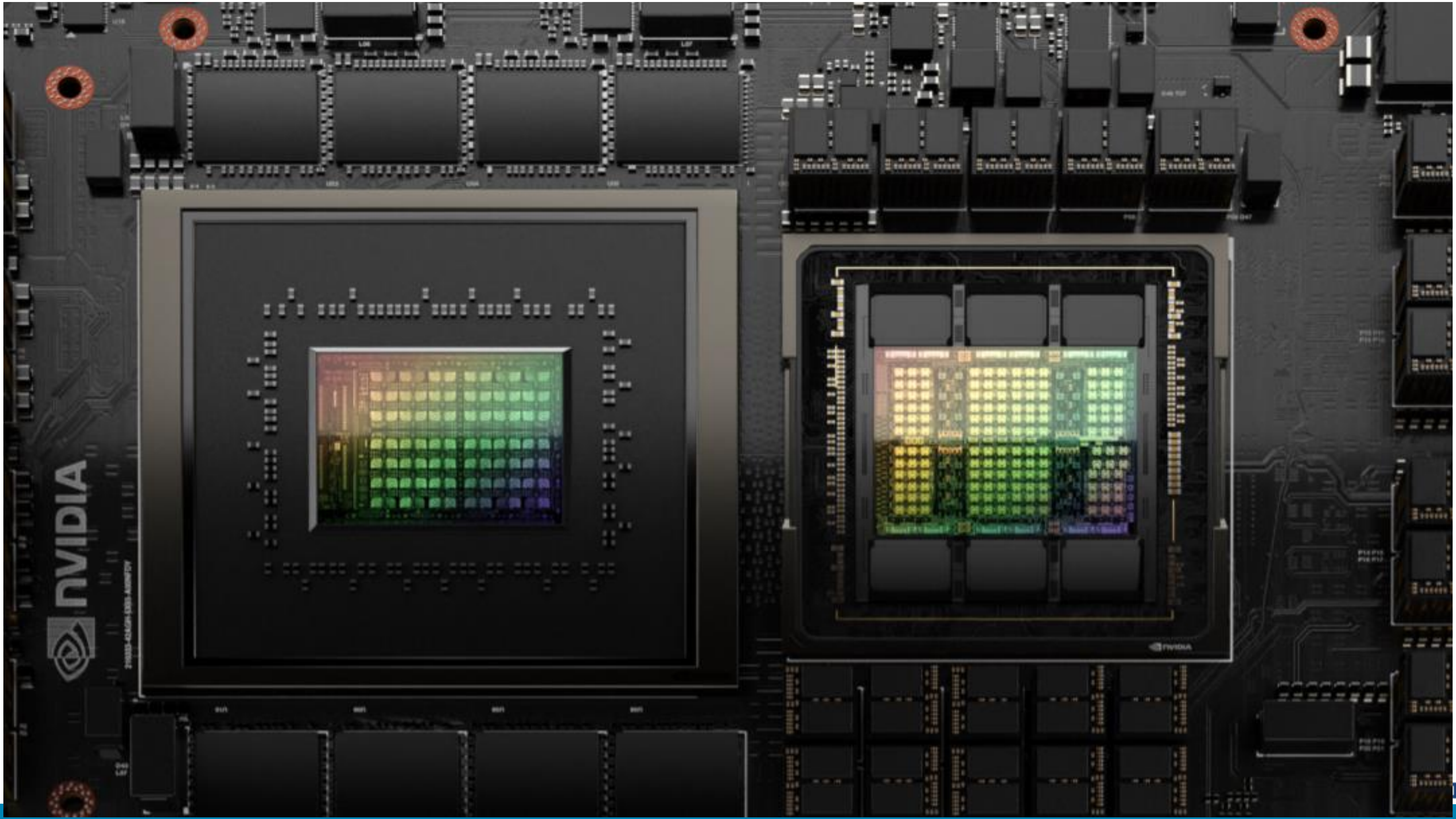
NVIDIA GH200 Grace-Hopper Superchip

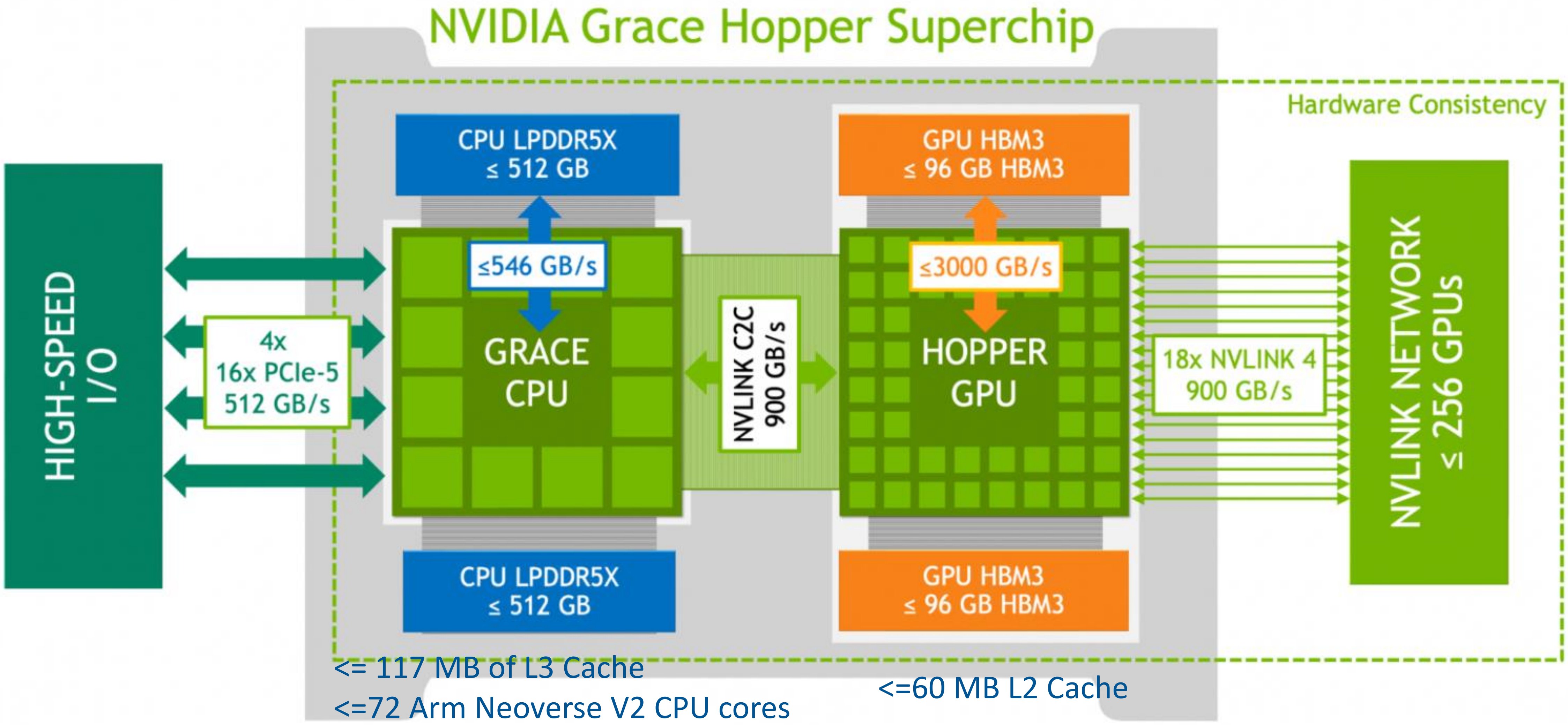


NVIDIA Grace Hopper Superchip

Brings together the performance of the NVIDIA Hopper GPU with the versatility of the ARM-baed NVIDIA Grace CPU.

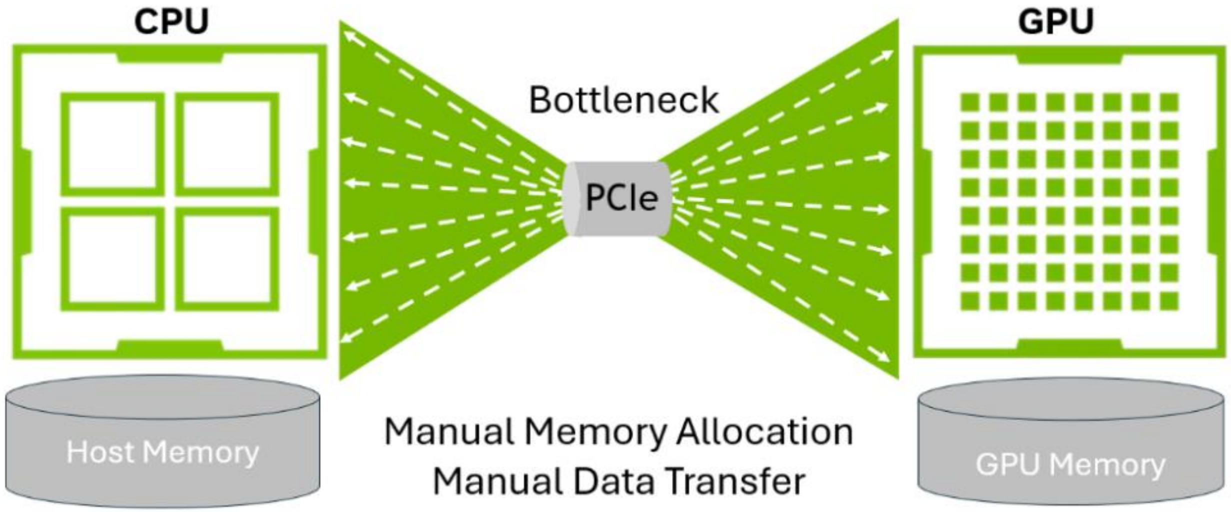
CPU and GPU are connected with a high bandwidth and memory coherent NVIDIA NVLink Chip-2-Chip (C2C) interconnect in a single superchip

It also supports NVIDIA NVLink Switch System.

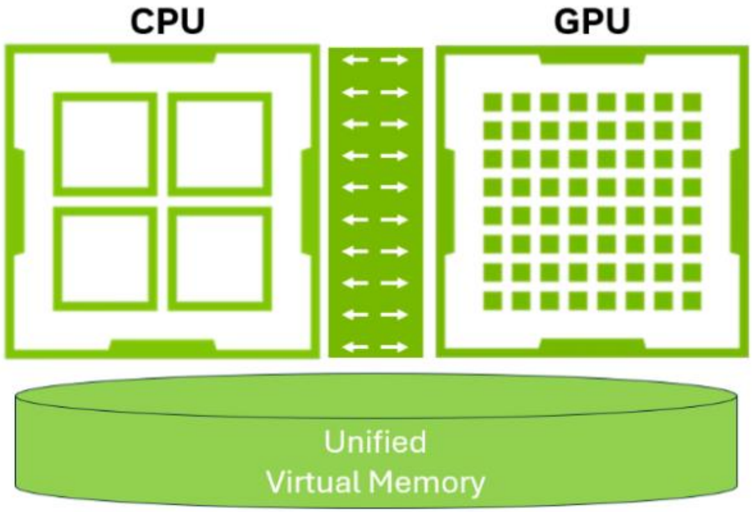


900GB is 7x higher bandwidth than x16 PCIe Gen5 lanes (128GB/s) commonly used in accelerated systems.
900GB/s means 450 GB/s per direction.

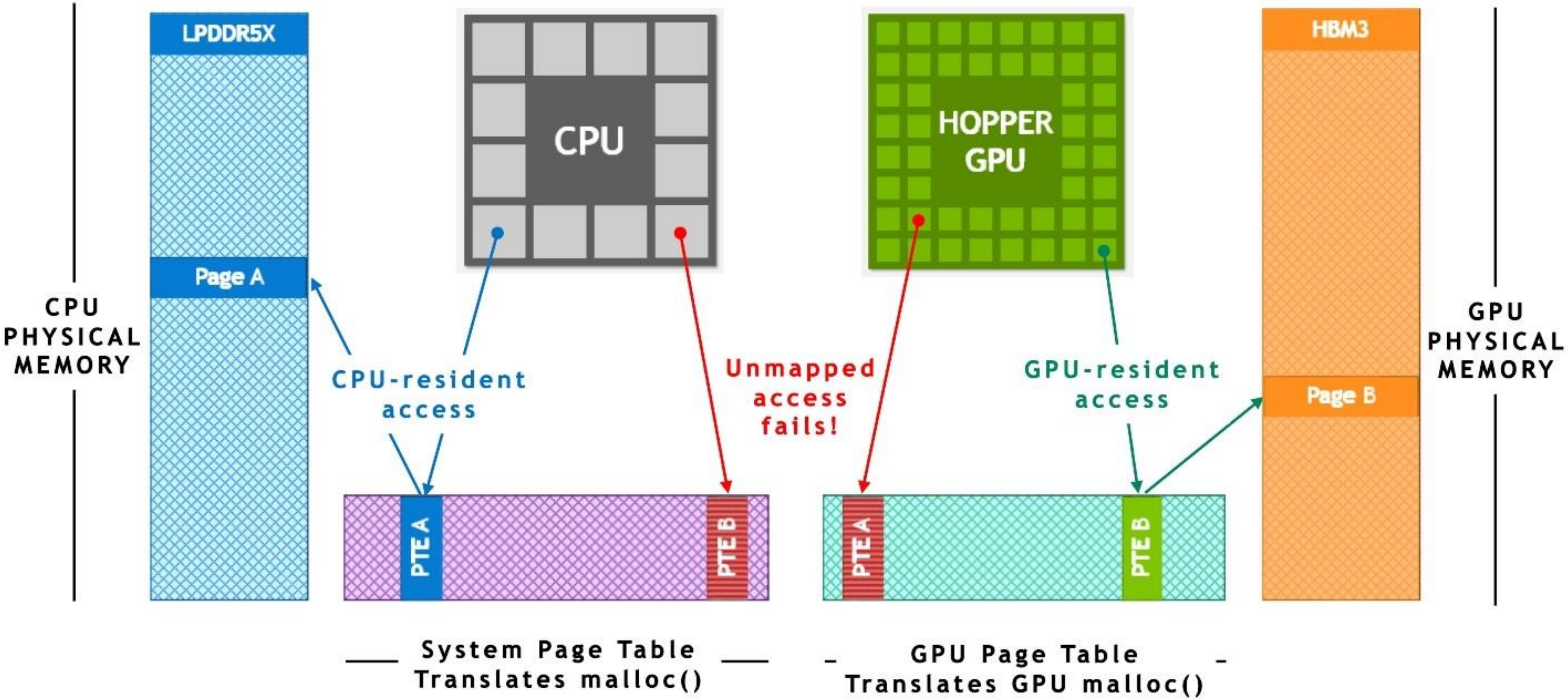
Legacy PCIe Architecture



Grace Hopper Architecture



Traditional PCIe-connected x86+Hopper systems without ATS



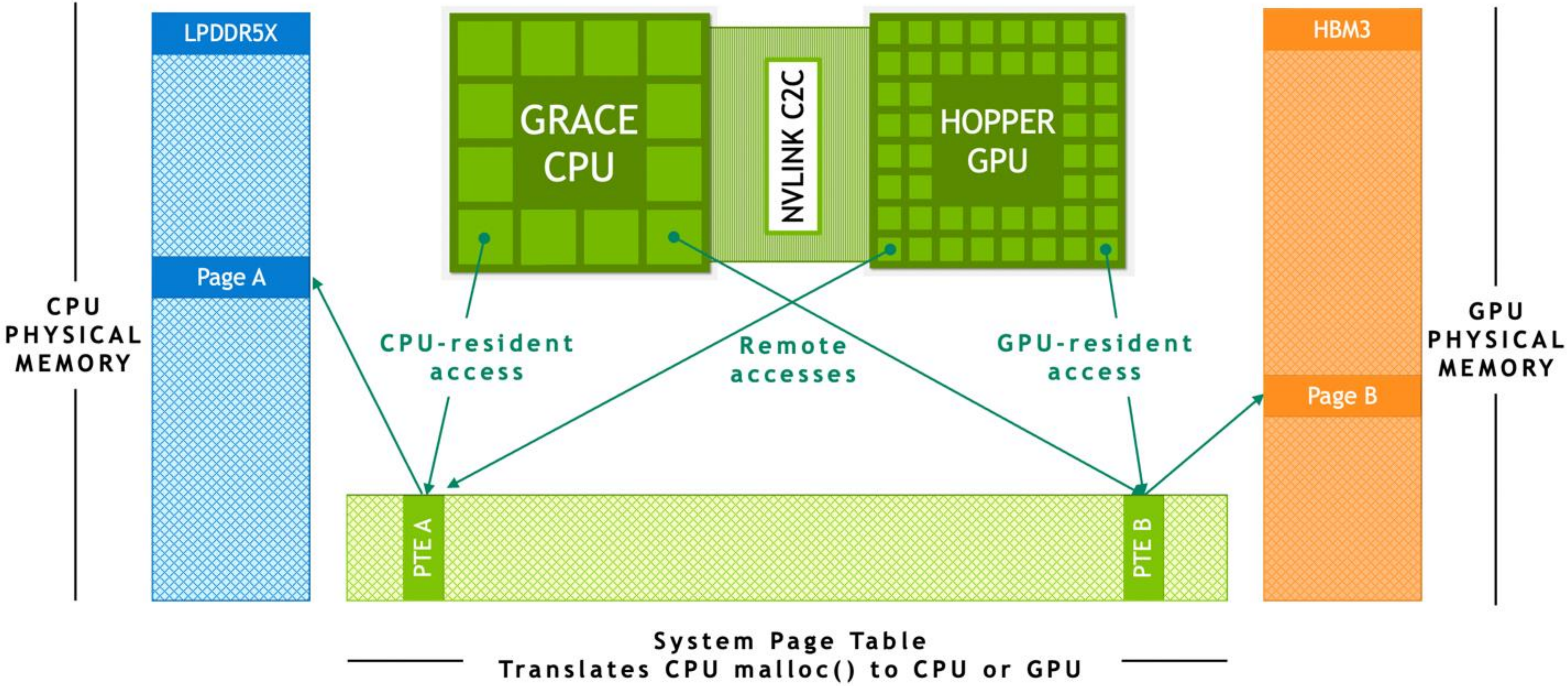
Traditional PCIe-connected x86+Hopper systems without ATS

The CPU and the GPU have independent per-process page tables

System-allocated memory is not directly accessible from the GPU.

When a program allocates memory with the system allocator but the page entry is not available in the GPU's page table, then accessing the memory from a GPU thread fails.

Grace Hopper Superchip with ATS



Grace Hopper Superchip with ATS

ATS enables the CPU and GPU to share a single per-process page table

This enables all CPU and GPU threads to access all system-allocated memory, which can reside on physical CPU or GPU memory.

The CPU heap, CPU thread stack, global variables, memory-mapped files, and interprocess memory are accessible to all CPU and GPU threads.

Advantages

It enables all CPU and GPU threads to access all system-allocated memory that can reside on physical CPU or GPU memory.

Each Hopper GPU can address up to 608 GB of memory within a superchip (512+96GB).

This architecture removes the need to copy memory back and forth between the CPU and GPU

NVLink-C2C

NVIDIA NVLink-C2C is an NVIDIA memory coherent superchip interconnect.

NVLink-C2C memory coherency enables GPUs to access large amounts of memory.

CPU and GPU threads can now concurrently and transparently access both CPU– and GPU-resident memory, enabling you to focus on algorithms instead of explicit memory management.

With memory coherency, one needs to transfer only the required data, and not migrate entire pages to and from the GPU.

Advantages

NVLink-C2C with Address Translation Services (ATS) uses Hopper's DMA copy engines for accelerating bulk transfers of pageable memory across host and device.

NVLink-C2C enables applications to oversubscribe the GPU’s memory and directly utilize NVIDIA Grace CPU’s memory at high bandwidth.

With up to 512 GB of LPDDR5X CPU memory per Grace Hopper Superchip, the GPU has direct high-bandwidth access to 4x more memory than what is available with HBM.

NVLink-C2C only uses 1.3 picojoules per bit transferred, which is greater than 5x more energy-efficient than PCIe Gen 5.

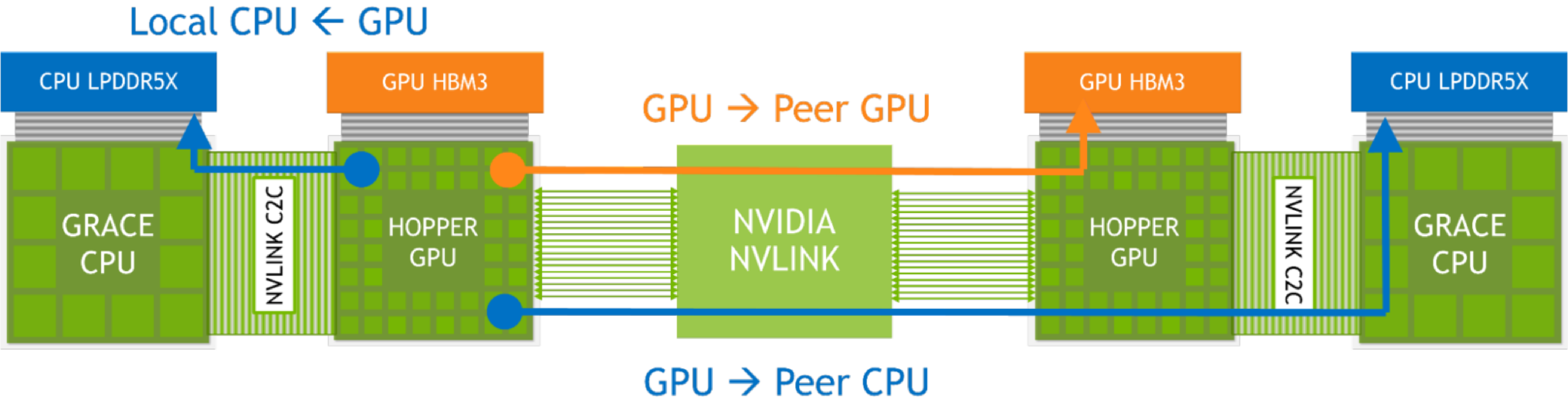
Scaling to more Superchips

Combined with the NVIDIA NVLink Switch System, all GPU threads running on up to 256 NVLink-connected GPUs can now access up to 150 TB of memory at high bandwidth.

Each NVLink-connected Hopper GPU can address all HBM3 and LPDDR5X memory of all superchips in the network

NVIDIA DGX GH200 is the first supercomputer to break the 100 terabyte barrier for memory accessible to GPUs over NVLink.

Memory accesses across NVLink-connected Grace Hopper Superchips



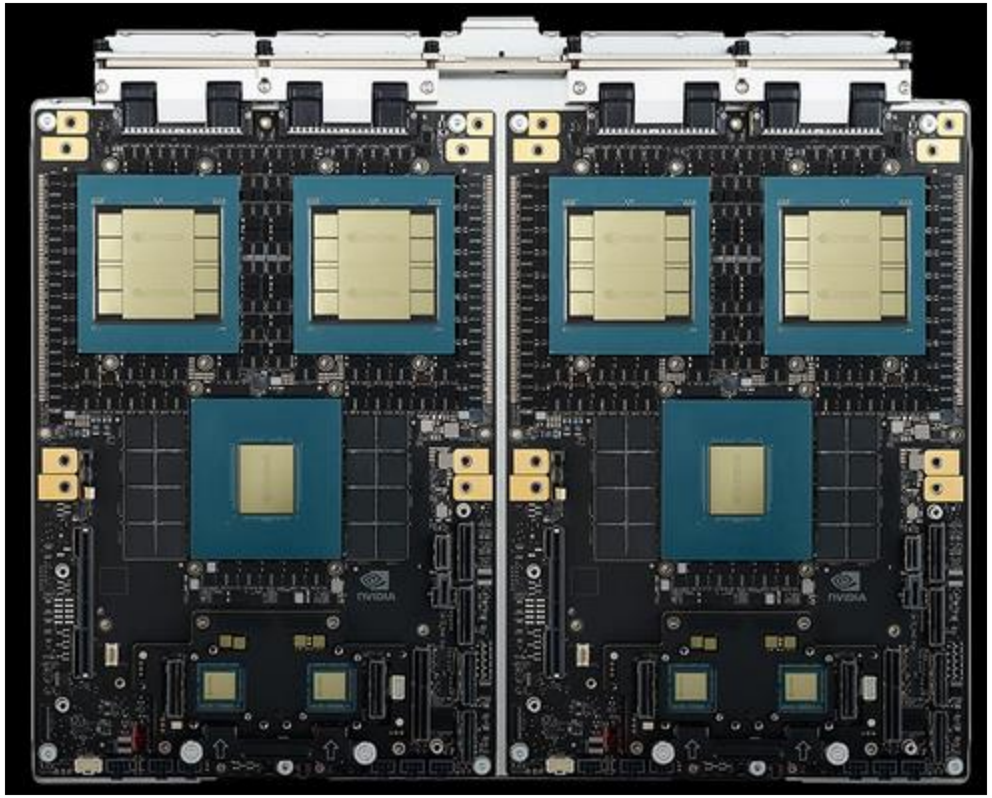
GPU threads can now access all memory resources available over the NVSwitch fabric, both LPDDR5X and HBM3, at 450 GB/s.

Speedup Grace Hopper vs x86+Hopper
5
4
3
2
1
4.0 x
3.5 x
1.9x
4.4x
3.6x
2.69x
1.3x
NLP
DLRM
GNN
Hash Join
ABINIT
OpenFOAM
GROMACS
ML Training
Databases
HPC

IIT ROORKEE

GB200

GB200 Grace-Blackwell superchip:
two Blackwell B200 GPUs and one Grace CPU



GB200 NVL4: four NVIDIA NVLink Blackwell GPUs unified with two Grace CPUs over NVLink-C2C interconnect

GB200 Grace Blackwell Superchip

	GB200 Grace Blackwell Superchip
Configuration	1 Grace CPU : 2 Blackwell GPU
FP4 Tensor Core[1]	40 | 20 PFLOPS
FP8/FP6 Tensor Core[2]	20 PFLOPS
INT8 Tensor Core[2]	20 POPS
FP16/BF16 Tensor Core[2]	10 PFLOPS
TF32 Tensor Core[2]	5 PFLOPS
FP32	160 TFLOPS
FP64 / FP64 Tensor Core	80 TFLOPS
GPU Memory | Bandwidth	Up to 372 GB HBM3e | 16 TB/s
NVLink Bandwidth	3.6 TB/s
CPU Core Count	72 Arm Neoverse V2 cores
CPU Memory | Bandwidth	Up to 480 GB LPDDR5X | Up to 512 GB/s

Each GPU has 192GB of HBM3e memory

CPU has 512GB of LPDDR5 memory through 16 memory channels.

This results in a total of 896GB of unified memory, accessible to all three devices.

The GB200 can scale up to 512 GPUs within a single NVLINK domain.

The primary building block for this configuration is the NVL72, which integrates 72 GPUs.

NVLINK is faster that an InfiniBand cluster

NVIDIA GB200 NVL72

	GB200 NVL72	GB200 Grace Blackwell Superchip
Configuration	36 Grace CPU : 72 Blackwell GPUs	1 Grace CPU : 2 Blackwell GPU
FP4 Tensor Core[1]	1,440 PFLOPS	40 | 20 PFLOPS
FP8/FP6 Tensor Core[2]	720 PFLOPS	20 PFLOPS
INT8 Tensor Core[2]	720 POPS	20 POPS
FP16/BF16 Tensor Core[2]	360 PFLOPS	10 PFLOPS
TF32 Tensor Core[2]	180 PFLOPS	5 PFLOPS
FP32	5,760 TFLOPS	160 TFLOPS
FP64 / FP64 Tensor Core	2,880 TFLOPS	80 TFLOPS
GPU Memory | Bandwidth	Up to 13.4 TB HBM3e | 576 TB/s	Up to 372 GB HBM3e | 16 TB/s
NVLink Bandwidth	130 TB/s	3.6 TB/s
CPU Core Count	2,592 Arm® Neoverse V2 cores	72 Arm Neoverse V2 cores
CPU Memory | Bandwidth	17 TB LPDDR5X | 14 TB/s	Up to 480 GB LPDDR5X | Up to 512 GB/s

Liquid Cooling

Liquid-cooled GB200 NVL72 racks reduce a data center's carbon footprint and energy consumption.

Liquid cooling increases compute density, reduces the amount of floor space used, and facilitates high-bandwidth, low-latency GPU communication with large NVLink domain architectures.

Compared to NVIDIA H100 air-cooled infrastructure, GB200 delivers 25x more performance at the same power, while reducing water consumption.

References

https://developer.nvidia.com/blog/nvidia-grace-hopper-superchip-architecture-in-depth/